

Adam Tang

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Available Spring + Summer 2027 (up to 8 months, co-op or internship)

Engineering Portfolio (CAD, builds, test videos): antangled.github.io/engineering_portfolio

EDUCATION

University of California, Berkeley

Expected June 2028

B.S. Electrical Engineering and Computer Sciences, Minor in Bioengineering | GPA: 3.91/4.00

Berkeley, CA

- **Honors:** Tau Beta Pi and Eta Kappa Nu (IEEE-HKN) engineering honor societies, Fall 2026

EXPERIENCE

Mechatronics and Embedded Systems Intern

Dec. 2025 – Aug. 2026

Eldaeon (RF sensing startup)

Berkeley, CA

- Built a drone and UAP detection pipeline in Python on Raspberry Pi with SDRs that cross-correlates RF Doppler returns with aircraft ADS-B and ship AIS tracks, flagging uncorrelated targets live with AWS and Twilio phone alerts
- Designed the antenna pan-tilt mechanism and wrote calibration firmware that sweeps $\pm 10^\circ$ to lock onto the strongest local transmitter, yielding the clearest Doppler signal; field-tested across 3+ outdoor campaigns and an SF boat-detection run
- Mapped each component's power budget from datasheets and wrote a power-sequencing daemon (DPOL) that keeps servo moves and SDR bursts from overlapping, cutting projected draw 30.2% to under the 15.4 W PoE limit

Solar Engineering Project Manager

Sep. 2025 – May 2026

CalSol, UC Berkeley Solar Racing Team

Berkeley, CA

- Designed the solar array and bypass-diode layout so the battery keeps charging even when panels are partially shaded, a projected +12% partial-shade efficiency gain
- Designed MPPT charge controller enclosures in SolidWorks and integrated them into the wiring layout with the batteries

Mechanical Lead

Nov. 2021 – June 2024

MUREX Exeter, MATE Marine Robotics Competition Team

Exeter, NH

- Led a 6-person team building a 6-DOF underwater ROV and 7-DOF manipulator; cut manufacturing cost 88% by moving outsourced parts to in-house CNC machining and 3D printing; ran design reviews; 6th globally at MATE 2024, 10 records

PROJECTS

Kinema: 6-DOF Camera Arm, Custom Cycloidal Actuators | *Self-initiated, SolidWorks, FEA* June 2026 – Present

- Designed a 10:1 internal cycloidal actuator packaged coaxially inside a frameless BLDC (10.4 N·m peak) by running tolerance stack-ups on the cycloid and pin fits, selecting bearings, and validating the housing with FEA
- Built 7+ CNC-machined and 3D-printed prototypes, then brought up the motor driver and absolute encoder for closed-loop position control; bench-tested to 2 N·m, earning a CubeMars sponsorship and \$1.5k Jacobs Ignite grant

PolySynth X: MPE Keyboard and Synth | *Self-initiated, STM32H7, C++, KiCad*

Nov. 2025 – June 2026

- Built a working 12-key MPE keyboard with an onboard synth engine by iterating compliant hammer-action keys 7+ times in Onshape and 3D print for touch accuracy, and designing hall-effect sensor PCBs in KiCad over 3+ revisions
- Wrote real-time C++ DSP firmware on STM32H7 (Daisy Seed) that differences two hall-effect sensors under each key via ADC, giving every key independent pitch bend

Modular Biodiversity Sensing Module (Hardware Lead) | *ESP32-S3, Solar Power, Edge AI*

Oct. 2025 – June 2026

- Designed a weatherproof, fully solar-powered sensor node (buck-converter power system, camera, acoustic, and thermal sensors) running an on-device AI model that identifies species with confidence scores; first node deployed in the field

Compact Automated Aquaponics System | *Self-initiated, Fusion 360, Sensors*

Dec. 2023 – June 2025

- Designed and hand-built an aquaponics system with a sensor and control stack (pH, ammonia, temperature, automated feeding) that monitors fish and plant health; deployed in the school dining hall as a hands-on teaching tool, still running

LEADERSHIP

Technical Project Manager and Systems Modeling Engineer

Feb. 2026 – Present

BEACN (Bay Area Environmentally Aware Consulting Network)

Berkeley, CA

- Lead a 5-person team advising GridPoint on battery storage for a multi-hundred-site rollout; benchmarked cell chemistries

Founder and President

Sep. 2020 – June 2025

Innovation Towards the Future (climate NGO)

Potomac, MD

- Founded and led a climate NGO for 5 years, earning admission as a UNFCCC observer organization and securing 3+ corporate partnerships; selected as USA youth delegate to UN COP28 in Dubai from 5k applicant NGOs

TECHNICAL SKILLS

Mechanical: SolidWorks, Fusion 360, Onshape, FEA, tolerance stack-ups, actuator and compliant mechanism design, DFM/DFA

Electrical: KiCad PCB design, board bring-up and validation, analog signal conditioning, BLDC drivers, encoders, SDR, MPPT

Embedded and Software: C++, Python, Java, Verilog, STM32H7, ESP32-S3, Raspberry Pi, ADC, real-time DSP, AWS, Twilio

Fabrication and Test: Rapid prototyping, CNC machining, 3D printing, assembly, bench and field testing, power budgeting